Energy 304 (2024) 132118

Contents lists available at ScienceDirect

Energy

journal homepage: www.elsevier.com/locate/energy

Syngas production through CO_2-mediated pyrolysis of polyoxymethylene

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ARTICLE INFO

Keywords:
Circular economy
Waste valorization
Pyrolysis
Carbon dioxide
Polyoxymethylene

ABSTRACT

Plastics have become an integral part of our daily lives owing to their exceptional physicochemical properties, such as durability, low density, and cost-effectiveness, compared to traditional materials. However, the escalating production of plastics has resulted in a proportional increase in waste generation. This paper proposes environmentally benign valorization/disposal methods for plastic waste, with a particular focus on adopting a pyrolysis process that utilizes CO_2 as a strategic reaction medium. As a case study, polyoxymethylene (POM), a widely used engineering plastic, was valorized through CO_2-mediated pyrolysis. This study experimentally demonstrates the mechanistic effectiveness of CO_2 in expediting the reaction kinetics of the thermal decomposition, specifically dehydrogenation and deoxygenation, of volatile matter derived from POM. The results revealed that employing CO_2 as a reactant in the two-stage pyrolysis at 500 °C produced 30.47 mmol more syngas than under inert conditions. In conclusion, the strategic utilization of a two-stage pyrolysis process at 500 °C with CO_2 as the reactant has emerged as an effective approach to the valorization of POM. This study contributes to developing sustainable methods for managing plastic waste by addressing environmental concerns and the need for efficient material recovery.

1. Introduction

Plastics play an indispensable role in a wide range of applications because of their superior properties such as high durability [1], low density [2,3], and low production cost [4] compared to traditional materials, such as wood, metals, and ceramics. Plastic versatility can be altered by incorporating additives, including chemicals (plasticizers and stabilizers) and materials (inorganics and glass fibers) [5]. Therefore, the demand for plastics has steadily increased, resulting in a 230-fold increase in global production over the past five decades (from 2 Mt in 1950 to 368 Mt in 2019) [6,7]. However, the surge in plastic production has led to a proportional increase in waste generation. The resulting plastic waste exhibits persistent characteristics given its extremely low biodegradability, raising environmental concerns [8]. Specifically, plastic waste exposed to the environment undergoes complex weathering processes, leading to its fragmentation into smaller particles known as microplastics (MPs) [9]. MPs possess a high capacity for adsorbing various pollutants, making them potential carriers for long-range pollutant transportation [10]. The convoluted fates of MPs highlight the need to curb the discharge of plastic waste into the environment.

Landfilling has traditionally been a method for sequestering plastic waste [11–13]. However, it is not considered a comprehensive solution because of the prolonged degradation time of plastic waste arising from its exceptionally low biodegradability [14]. Another approach involves incineration, aimed at volume reduction and energy recovery [15]. However, incineration is not environmentally benign, as it generates air pollutants such as CO, particulates (soot), and unburned hydrocarbons through incomplete combustion [16]. Their formations underscore the need for environmentally benign platforms for plastic waste disposal. Simultaneously, the circular economy concept has gained attention as part of the United Nations 2030 agenda on sustainable development goals [17]. In alignment with this, plastic waste recycling has been actively pursued.

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https://doi.org/10.1016/j.energy.2024.132118
Received 22 February 2024; Received in revised form 31 May 2024; Accepted 16 June 2024
Available online 17 June 2024
0360-5442/

Plastics consist of repeating monomers, leading to the depolymerization of plastic waste into monomers via chemolysis [18]. However, the high crystallinity of several plastic types such as polyoxymethylene (POM), polyethylene, and polypropylene poses technical challenges in recovering monomers from plastic waste [19]. Thus, chemolysis cannot be applied universally to recylce plastic waste. Pyrolysis can be a desirable alternative for recycling plastic waste [20,21]. Pyrolysis is a thermally destructive process that transforms carbonaceous materials into smaller molecules in the absence of oxygen, resulting in liquid, gas, and solid-phase pyrogenic products [22–24]. While plastic waste is commonly pyrolyzed to obtain a liquid product (oil) [25,26], the complex nature of pyrolysis oil poses a considerable challenge in its direct use as fuel [27,28]. Consequently, additional upgrading processes—distillation and reforming—are imperative, deteriorating economic feasibility [29]. Therefore, converting plastic waste into gaseous products, particularly syngas, has emerged as a beneficial option. Syngas—comprising H_2 and CO—is versatile and can be directly utilized as fuel or chemical building blocks [30,31].

POM is a major engineering plastic widely used in precision parts that require high stiffness, less friction, and excellent dimensional stability [32,33]. In 2021, the global market size of POM will reach 5.1 billion USD and an estimated annual growth rate of 5.5 % from 2022 to 2028 [19], and the generation of end-of-life POM is expected to increase. Notably, POM is composed of the potentially toxic formaldehyde monomers, which can be easily released into the environment during mechanically abraded or exposure to heat and solar radiation [34]. As such, it is urgent to secure a proper method for the disposal of POM. Based on these rationales, this study proposes an environmentally benign platform that disposes of POM with its valorization into syngas through a pyrolysis process. CO_2, one of the greenhouse gases, was intentionally employed as the reaction medium in the pyrolysis of POM. Indeed, CO_2 has been utilized as a reactive gas agent to control the fate of pollutants and increase the generation of syngas in the gasification process of plastic waste [35]. However, this process is energy-intensive considering its operational temperature (700–1000 °C) [36]. Thus, this study emphasized understanding of the mechanistic reactivity of CO_2 as a reaction medium at the moderate temperature (≤700 °C) in the pyrolysis process. A fundamental study of the role of CO_2 in the pyrolysis of POM involves thermogravimetric investigations of the thermal behavior of POM under CO_2. Subsequently, POM was pyrolyzed under CO_2, and the resulting effluents, including liquid and gases, were analyzed quantitatively and qualitatively. For reference, all experiments related to the pyrolysis of POM under CO_2 were conducted under N_2.

2. Materials and methods

2.1. Sample and chemical reagents

The POM samples used here were sourced from Kolon Plastics (Korea). Dichloromethane (CH_2Cl_2, 99.5 %) was acquired from Daejung Chemical (Korea). Ultra-high purity gases (N_2 and CO_2, 99.999 %) were supplied by Green Gas (Korea). The calibration gas mixture comprising H_2, N_2, CH_4, CO, CO_2, C_2H_6, C_2H_4, and C_2H_2 was obtained from INFICON (Switzerland).

2.2. POM characterization

POM identification involved Fourier-transform infrared (FT-IR) and ultimate analyses. The FT-IR analysis utilized an Agilent Cary 630 instrument (Agilent Technologies, USA), while the ultimate analysis was conducted using a FlashSmart™ elemental analyzer (Thermo-Fisher Scientific, USA).Thermogravimetric analysis (TGA) was performed to examine the POM thermal characteristics in N_2 and CO_2 conditions. The TGA experiments were conducted with a 10 ± 0.01 mg sample loading using a STA449 F5 Jupiter instrument (NETZSCH, Germany). The total flow rates, including purge gas (N_2 and CO_2) at 80 mL min^{-1} and protective gas (N_2) at 20 mL min^{-1}, were adjusted to 100 mL min^{-1}. The temperature program was set from 40 to 900 °C with a heating rate of 10 °C min^{-1}. The heating rate was identically maintained for pyrolysis studies to understand the relationships between thermal characteristics of POM and initiation temperature for the production of pyrogenic liquid and gases. TGA tests were conducted at least three times. The standard deviation of the experiments was less than 2 %. For better display, we did not add errors in Figures. Prior to conducting TGA experiments, the temperature and sensitivity of the simultaneous thermal analysis (STA) instrument were calibrated using the standard materials (In, Sn, Zn, Al, Ag, and Au).

2.3. Pyrolysis setup

To construct the pyrolysis reactors, a hollow quartz tube was attached to an external tubular furnace (FT-830; DAIHAN Scientific, Korea). For one-stage pyrolysis, one tubular furnace was equipped and operated from 100 to 610 °C at a heating rate of 10 °C min^{-1}. POM (5 ± 0.01 g) was placed in the center of the tubular furnace. Two ultra-torr vacuum fittings (Swagelok SS-4-UT-6-600, USA) were connected to both ends of the hollow quartz tube to facilitate connections between the inlet and outlet. The flow rates of N_2 and CO_2 (80 vol % N_2 balance) were set at 800 mL min^{-1}. A mass-flow controller (5850 Series E, Brooks Instruments, USA) adjusted the flow rate. For two-stage pyrolysis, an additional tubular furnace was placed alongside the first furnace. The first tubular furnace operated under the same temperature program (from 100 to 610 °C with a heating rate of 10 °C min^{-1}), and the second furnace was maintained at isothermal temperatures of 500 and 700 °C. The purpose of the two-stage pyrolysis was the additional thermal cracking of volatile matters (VMs) derived from POM (originated from pyrolysis in the first heating zone) through the second heating zone. All test temperatures were monitored using a data logger (GL840-M, Graphtec, Japan) connected with a K-type thermocouple (Omega, USA). The schematic layout of the pyrolysis setups (one-stage and two-stage pyrolysis) are shown in supplementary material (Fig. S1). All pyrolysis experiments were conducted three times. The standard deviations of the experiments were less than 2 %. Thus, they did not present in the figure for a better display.

2.4. Pyrogenic product characterization

The pyrogenic products resulting from the pyrolysis of POM underwent direct cooling at −40 °C using a chiller (WTC-40, DAIHAN Scientific, Korea). The condensed pyrogenic products, referred to as pyrogenic liquids (oil), were diluted with DCM and subsequently identified by gas chromatography-mass spectrometry (GC-MS) using a Clarus 500 instrument (PerkinElmer, USA). The GC-MS instrument was equipped with an Agilent DB-WAX column (30 m × 0.25 mm × 0.25 μm). NIST mass spectral library (version 2020) was used to identify the chemical compounds in condensed pyrogenic products. The uncondensed pyrogenic products, known as pyrogenic gases, were directly introduced into a Micro-GC (3000A, INFICON, Switzerland). This Micro-GC was equipped with a thermal conductivity detector featuring a Molecular sieve column (10 m × 320 μm × 30 μm) and Plot U column (8 m × 320 μm × 30 μm). Prior to quantifying the major pyrogenic gases (H_2, CH_4, and CO), the micro-GC was calibrated using a INFICON gas mixture (comprising H_2, N_2, CH_4, CO, CO_2, C_2H_6, C_2H_4, and C_2H_2). The overall experimental flow diagram in this study is shown in supplementary material (Fig. S2).

3. Results and discussion

3.1. POM identification

Commercial POM can be classified into two types based on the number of monomers [37]. POM homopolymers are composed of a single monomer (formaldehyde), and POM copolymers comprise two monomers (formaldehyde and cyclic ethers, such as ethylene oxide) [38]. These distinctions in monomer composition suggest inherent differences in their structural properties. The thermolytic behavior of polymers is closely linked to their structural characteristics [39]. Consequently, identifying the specific POM types is crucial for fundamental POM pyrolysis studies. In this context, POM was determined through FT-IR analysis, recognized as an effective tool for the identification of different plastic types [40]. FT-IR analysis provides a spectrum that serves as a molecular fingerprint, aiding in the characterization of the polymer types [41]. The FT-IR spectrum obtained from the POM analysis is presented in Fig. 1.

The POM FT-IR spectrum exhibited six characteristic peaks related to CH stretching vibrations (2972, 2924, and 2844 cm^{-1}), CH_2 rocking and C–O–C bending vibrations (1214 cm^{-1}), and C–O–C antisymmetric stretching and CH_2 rocking vibrations (1079 and 887 cm^{-1}). These peaks are characteristic and can be identified in both POM homopolymers and copolymers [42,43]. To distinguish between POM copolymers and homopolymers, previous studies have indicated that the intensity of the two peaks related to CH_2 and CH_3 bending vibrations (1380 and 1480 cm^{-1}) can be crucial [37,44]. Specifically, these peaks are representative of ethylene oxide in the POM copolymers, with higher intensities observed in the POM copolymers than in the homopolymers. However, in this study, the identification of the POM type based on FT-IR analysis was challenging because of the absence of FT-IR results for the POM homopolymers as a reference. As an alternative approach, the ultimate analysis was employed to determine the elemental composition of POM, considering C, H, O, N, and S [45]. 40.33 wt % of C, 6.81 wt % of H, 52.1 wt % of O, 0.76 wt % of N were determined through the ultimate analysis. The results of ultimate analysis revealed higher moles of C and H in the POM empirical chemical formula ($C_{1.03}H_{2.09}O_{1.00}N_{0.02}$) compared to the expected chemical formula ($(CH_2O)_n$) of the POM homopolymer [43]. The higher C and H contents aligned with the expectations for POM copolymers, considering the addition of ethylene oxide (C_2H_4O). Therefore, it can be concluded that the POM used here is a copolymer. In addition, the presence of N content (0.76 wt %) suggests the inclusion of additives, such as antioxidants, formaldehyde scavengers, and other thermal stabilizers, in the POM sample [46].

3.2. POM thermal behavior under CO_2

To investigate the thermolytic behavior of POM under CO_2 conditions, a TGA test was conducted. A POM sample weighing 10 ± 0.1 mg underwent thermal decomposition in the temperature range of 40–900 °C, with a heating rate of 10 °C min^{-1}. For comparison, a TGA test under N_2 conditions was also performed. Except for the gas environment, all the TGA test parameters under N_2 conditions were identical to those under CO_2. The resulting mass changes and curves for the differential thermograms of POM under N_2 and CO_2 conditions are presented in Fig. 2.

As illustrated in Fig. 2, a substantial POM mass change (99.57 wt %) was observed from 300 to 423 °C, resulting in a negligible residual mass. This indicates that most carbons in POM were transformed into liquid and/or gas phase products, completing the process at ≤ 423 °C. Two distinctive peaks in the differential thermogram curve were noted at 347 and 379 °C. This observation is likely due to the two different monomer units, such as formaldehyde and ethylene oxide, in POM [39]. Specifically, the thermal degradation of POM copolymer initiates with the cleavage of C–O bonds within part of the polymeric backbones consisting of formaldehyde, leading to the complete unzipping. After unzipping is reached to the next copolymer unit, the process is terminated. To continue the unzipping process, a new initiation step is required, which leads to the deceleration of the degradation rates [39].

Under CO_2 conditions, the POM thermolytic pattern differed from that under inert (N_2) conditions. In comparison to the N_2 conditions, the mass change under CO_2 occurred within a narrower temperature range from 300 to 412 °C, with the maximum degradation rate observed at 352 °C. This observation suggests that CO_2 played a role in causing POM to decompose at lower temperatures. Notably, the melting point of the POM copolymer is 170 °C. Therefore, the observed phenomenon suggests the occurrence of a heterogeneous reaction between the liquid phase of POM and CO_2. Among potential reactions, the Boudouard reaction is a well-known CO_2-induced heterogeneous reaction, where CO_2 reacts with solid carbon to form CO [47], and it typically occurs spontaneously at temperatures ≥710 °C [48]. This suggests that the observed heterogeneous reaction is not attributable to the Boudouard reaction. Nevertheless, a complete understanding of this heterogeneous reaction is limited with only TGA results, which lack additional information regarding the evolved products. To address this issue, lab-scale pyrolysis of POM under N_2 and CO_2 conditions was conducted, and the gaseous and liquid products were qualitatively and quantitatively characterized.

3.3. One-stage POM pyrolysis under CO_2 conditions

To gain insight into the mechanistic roles of CO_2 in POM thermolysis, one-stage POM pyrolysis was conducted under both N_2 and CO_2 conditions. The pyrolysis was conducted in the temperature range of

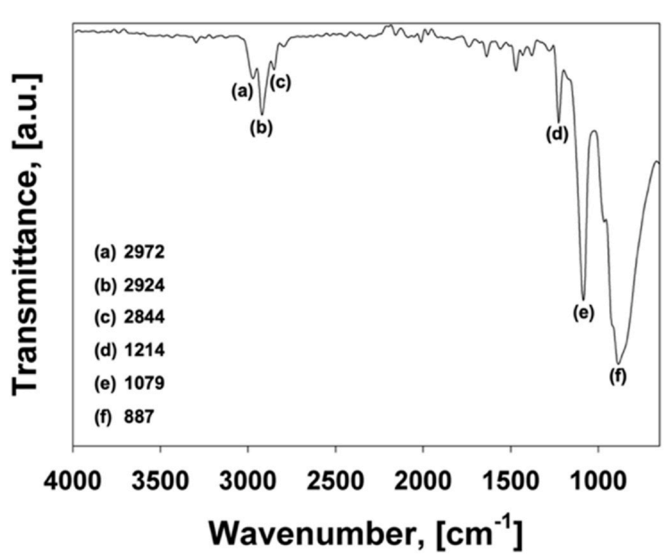

Fig. 1. POM FT-IR spectrum.

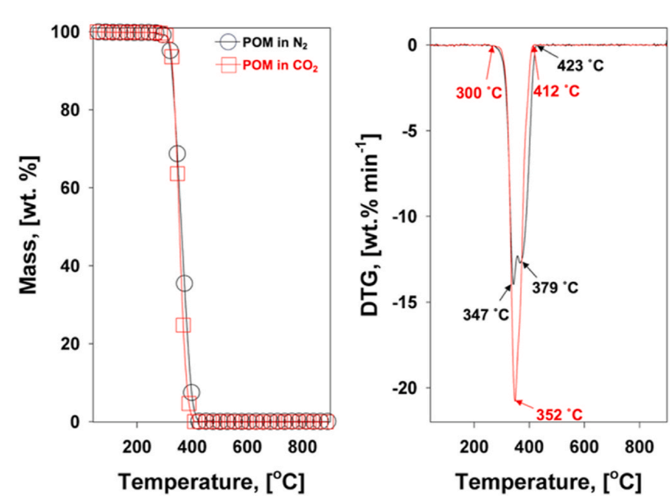

Fig. 2. POM mass changes and its differential thermogram curves under inert (N_2) and CO_2 conditions.

100–610 °C at a heating rate of 10 °C min^{-1}. The concentrations of the major gaseous products (H_2, CH_4, and CO) resulting from the one-stage POM pyrolysis under both conditions were monitored. The concentrations of these major gaseous products were then plotted as a function of the pyrolysis temperature, and their profiles are illustrated in Fig. 3.

Under N_2 conditions, the evolution of H_2, CH_4, and CO was observed from 310 to 430 °C, stemming from random bond scissions in the polymeric backbone of POM. Notably, the evolution of H_2 is lower than that of CO. This discrepancy can be attributed to the stronger bonding energy of C–H compared to C–C and C–O, as evidenced by the bonding energy values of C–C (347 kJ mol^{-1}), C–O (358 kJ mol^{-1}), and C–H (413 kJ mol^{-1}) [49]. Under CO_2 conditions, the evolution patterns of the major gaseous products differed from those under N_2. Specifically, increased H_2 and CO evolution was observed at temperatures up to 350 °C, followed by decreased evolution at temperatures ≥350 °C. This observation can be attributed to the heterogeneous reaction between CO_2 and the solid POM phase, with reaction kinetics maximized at 352 °C, as confirmed in Fig. 2. This heterogeneous reaction may modify the constituents of the liquid phase resulting from POM pyrolysis. To validate this, a semi-solid and liquid mixture that evolved from the POM pyrolysis under both conditions was characterized. The chromatograms of the liquid products and major chemical species under both conditions are shown in Fig. 4. All identified chemical species are summarized in Table 1.

Under N_2 conditions, alkyl ethers bearing methoxy and ethoxy groups resulting from POM pyrolysis were identified in the semi-solid and liquid mixture. In addition, a wide range of carbon lengths ($C_{3\text{-}14}$) were observed. This constitutional property of a semi-solid/liquid mixture is an important factor that hinders its practical use as a fuel. The resulting chromatograms of the semi-solid/liquid mixture and its chemical constituents under CO_2 were almost identical to those under N_2 (data not shown). This observation suggests that the heterogeneous reaction between CO_2 and POM did not modify the constituent properties of the semi-solid/liquid mixture. It is important to note that chemical species with higher carbon numbers are more challenging to analyze using GC because of their higher vapor pressure [50]. This implies that clues regarding the role of CO_2 in heterogeneous reactions may be observed in chemical species that cannot be analyzed using GC. Nonetheless, it was clear that CO_2 could expedite the thermolysis of the liquid phase of POM through a heterogeneous reaction, as shown in Figs. 2 and 3. Based on this result, it was assumed that the thermolysis of POM induced by CO_2 could be further accelerated through a homogeneous (gas-phase) reaction between CO_2 and the VMs produced during POM thermolysis. The effectiveness of CO_2 in homogeneous reactions with VMs from lignocellulosic biomass is initiated at 500 °C [51]. Taking this into consideration, the two-stage POM pyrolysis under CO_2 conditions was conducted. Specifically, the VMs that evolved from POM thermolysis in the first heating zone were directed to the second heating zone, providing an environment for a homogeneous reaction between CO_2 and VMs.

3.4. Two-stage POM pyrolysis under CO_2 conditions

To explore the CO_2-induced homogeneous reaction, two-stage POM pyrolysis under CO_2 conditions was performed. For this purpose, an additional furnace was added next to the original furnace and operated isothermally at 500 °C. This choice of isothermal temperature was influenced by the initial temperature reported in our previous study of CO_2-induced homogeneous reactions [51]. The major gaseous products that evolved from the two-stage POM pyrolysis under CO_2 conditions were monitored, and their resulting concentration profiles are presented in Fig. 5.

Under N_2 conditions, the formation of major pyrogenic gases was enhanced compared to that from one-stage pyrolysis. This enhancement can be attributed to the additional thermal cracking involving dehydrogenation and deoxygenation of VMs from POM at 500 °C. Under CO_2 conditions, the formations of H_2 and CO were further enhanced compared to those under N_2 conditions. CO_2 expedited the dehydrogenation and deoxygenation of VMs from POM. However, the detailed CO_2 role in homogeneous reaction cannot be elucidated in this stage due to lack of information. For a more in-depth study of the mechanistic effect of CO_2, changes in the pyrogenic liquid under N_2 and CO_2 conditions were also monitored, as illustrated in Fig. 6(a) and summarized in Table 2.

Under N_2 conditions, the chemicals in the pyrogenic liquid obtained from the two-stage pyrolysis at 500 °C were nearly similar to the results from the one-stage pyrolysis. However, the overall peak areas were notably reduced during the two-stage pyrolysis process. Considering the enhanced formation of H_2 and CO from the two-stage pyrolysis at 500 °C (Fig. 5), this observation indicates that a higher temperature contributes to the conversion of pyrogenic liquid to pyrogenic gases. Interestingly, the chemical composition matrix of the pyrogenic liquid from the two-stage pyrolysis at 500 °C was markedly changed under CO_2 conditions. Specifically, the formation of oxygen-containing hydrocarbons decreased and that of aromatic hydrocarbons considerably increased under CO_2 conditions. Notably, 1,2,3,4,5,6-hexamethylbenzene exhibited the largest peak area. As confirmed, chemical species in pyrogenic liquid under N_2 conditions includes ethylene oxide monomer. This infers that the deoxygenation of VMs should be prioritized to form the hydrocarbon (HC) intermediates. The HC intermediates are vulnerable to

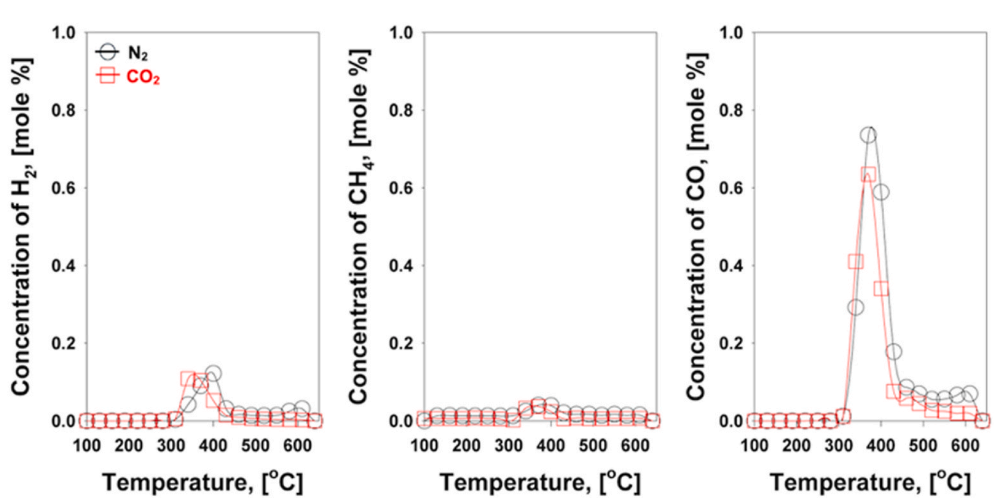

Fig. 3. Concentration profiles of the major pyrogenic gases resulting from one-stage POM pyrolysis under N_2 and CO_2 conditions.

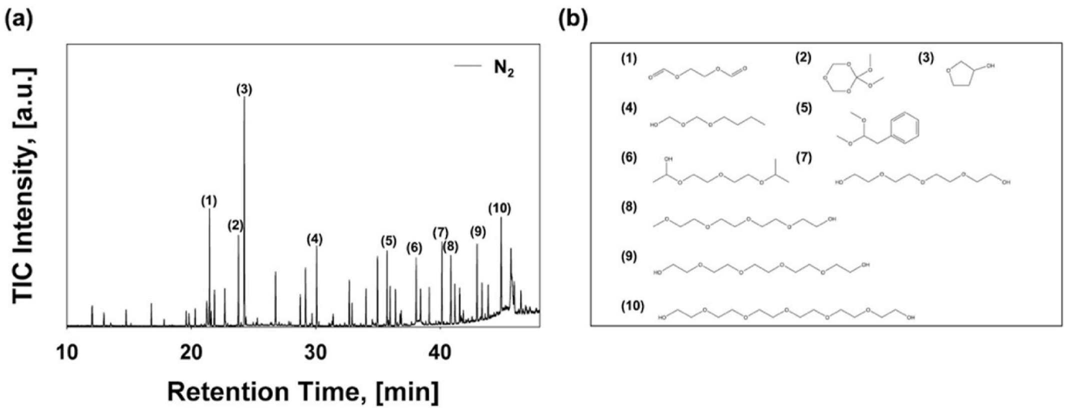

Fig. 4. (a) Chromatogram of pyrogenic liquid and (b) major chemical compounds identified in pyrogenic liquid obtained from one-stage POM pyrolysis under N_2 conditions.

Table 1
Identification of the major chemical compounds in pyrogenic liquid obtained from one-stage POM pyrolysis under N_2 conditions.

No.	RT, [min]	Chemical Compounds	Formula	Peak Area ($\times 10^6$)
1	21.48	ethane-1,2-diyldiformate (1)	$C_4H_6O_4$	15.06
2	23.8	2,2-dimethoxy-1,3,5-trioxane (2)	$C_5H_{10}O_5$	11.74
3	24.28	tetrahydrofuran-3-ol (3)	$C_4H_8O_2$	32.74
4	26.79	1-(hydroperoxymethoxy)pentane	$C_6H_{14}O_3$	7.37
5	29.2	2-(2-(2-methoxyethoxy)ethoxy) ethan-1-ol	$C_7H_{16}O_4$	8.03
6	30.08	(butoxymethoxy)methanol (4)	$C_8H_{18}O_5$	10.31
7	32.71	2,5,8,11-tetraoxatridecan-13-ol	$C_9H_{20}H_5$	6.09
8	34.97	(2,2-dimethoxyethyl)benzene (5)	$C_{10}H_{14}O_2$	9.61
9	35.74	1-(2-(2-isopropoxyethoxy)ethoxy) ethan-1-ol (6)	$C_9H_{20}O_4$	9.8
10	38.09	1-(1-ehtoxyethoxy)-3-methylpent-2-ene	$C_{10}H_{20}O_2$	7.96
11	38.44	1-ethoxy-4-methoxybutane-2,3-diol	$C_7H_{16}O_4$	5.86
12	40.16	tetraethyleneglycol (7)	$C_8H_{18}O_5$	11.6
13	40.87	2,5,8,11-tetraoxatridecan-13-ol (8)	$C_8H_{18}O_5$	10.25
14	41.21	2,3-diisopropoxy-1,4-dioxane	$C_{10}H_{20}O_4$	5.83
15	42.98	3,6,9,12-tetraoxatetradecane-1,14-diol (9)	$C_{10}H_{22}O_6$	10.5
16	44.92	3,6,9,12,15-pentaoxanonadecane-1,17-diol (10)	$C_{14}H_{30}O_6$	14.99

cyclization and dehydrogenation to form romatic hydrocarbons [52]. In this context, it can be concluded that the CO_2 partially oxidizes carbon-bearing oxygen in chemical species by reducing itself into CO, leading to the deoxygenation of VMs. This infers that CO_2 contributes to the shifting carbon in pyrogenic liquid into gases, which demonstrated by the reduction of pyrogenic liquid under CO_2 conditions, confirmed from the result of overall mass balance in Fig. 6(b).

These observations do not fully agree with the results from our previous studies [53–55]. In specific, incorporating CO_2 in non-catalytic pyrolysis of plastics, such as PE, PP, PET, *etc.*, did not enhance syngas production. Such discrepant results infer that the reactivity of CO_2 for CO production through homogeneous reaction depends on the physico-chemical properties of volatile matters from thermolysis of POM. However, it is hard to elucidate this phenomenon in this study. Thus, the related studies have to be done in near future.

In specific, CO_2 was not reactive with solid plastics in heterogeneous reactions. Also, the enhanced formation of syngas was not observed from non-catalytic pyrolysis. Unlike plastic types used in previous studies, POM showed different thermolytic behavior in CO_2 conditions. In detail, the heterogeneous reaction between CO_2 and liquid phase of POM was exhibited in this study. This suggests that the reactivity of CO_2 is higher in POM than other plastics. Such higher reactivity of CO_2 has led to the enhanced formation of syngas through the non-catalytic pyrolysis through homogeneous reaction with volatile matters from thermolysis of POM. These results show that the physico-chemical properties of VMs from thermolysis of plastics is crucial factor in homogeneous reaction. However, it is hard to elucidate this phenomenon in this study. Thus, the

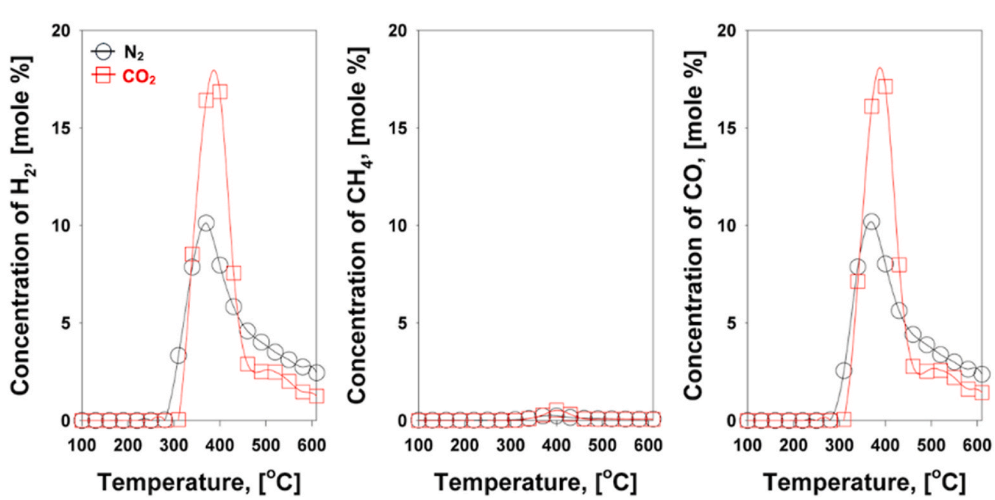

Fig. 5. Concentration profiles of the major pyrogenic gases resulting from two-stage POM pyrolysis (second furnace: 500 °C) under N_2 and CO_2 conditions.

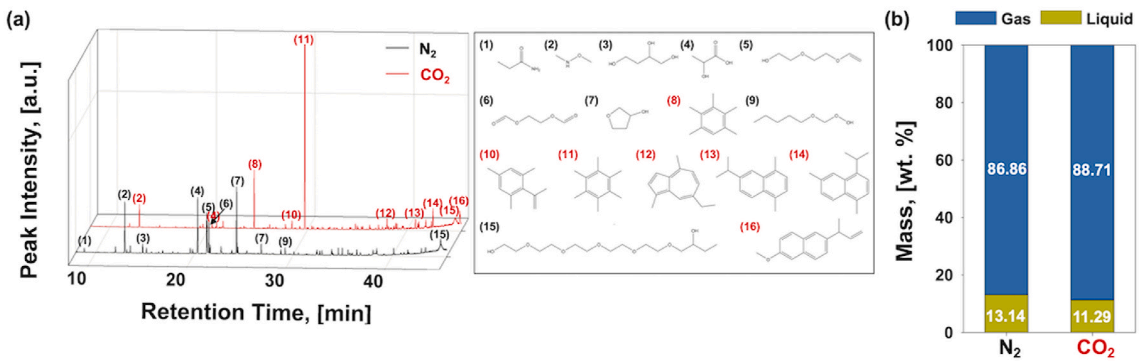

Fig. 6. (a) Chromatogram of pyrogenic liquid and major chemical compounds identified in pyrogenic liquid obtained from two-stage POM pyrolysis (second furnace: 500 °C) under N_2 and CO_2 conditions (b) overall mass balances of pyrogenic products from two-stage POM pyrolysis (second furnace: 500 °C) under N_2 and CO_2 conditions.

Table 2
Identification of the major chemical compounds in pyrogenic liquid obtained from two-stage POM pyrolysis (second furnace: 500 °C) under N_2 and CO_2 conditions.

No.	RT, [min]	Chemical Compounds	Formula	Peak Area	
				N_2	CO_2
1	8.88	2-hydroxypropanamide	$C_3H_7NO_2$	1.06	n.d.
2	12.97	Methoxymethylamine	C_5H_7NO	11.69	0.88
3	14.77	butane-1,2,4-triol	$C_4H_{10}O_3$	2.16	n.d.
4	20.32	2-hydroxypropanoic acid	$C_3H_6O_3$	12.42	1.18
5	21.27	2-(2-(vinyloxy)ethoxy)ethan-1-ol	$C_6H_{12}O_3$	12.10	n.d.
6	21.48	ethane-1,2-diyldiformate	$C_4H_6O_4$	6.33	1.54
7	24.28	tetrahydrofuran-3-ol	$C_4H_8O_2$	14.58	n.d.
8	24.58	1,2,3,4,5-pentamethylbenzene	$C_{11}H_{16}$	n.d.	12.24
9	26.79	1-(hydroperoxymethoxy) pentane	$C_6H_{14}O_3$	2.20	n.d.
10	28.40	1,3,5-trimethyl-2-(prop-1-en-2-yl)benzene	$C_{12}H_{16}$	n.d.	1.81
11	29.69	1,2,3,4,5,6-hexamethylbenzene	$C_{12}H_{18}$	n.d.	115.66
12	38.09	7-ethyl-1,4-dimethylazulene	$C_{14}H_{16}$	n.d.	3.06
13	40.88	6-isopropyl-1,4-dimethylnaphthalene	$C_{15}H_{18}$	n.d.	2.68
14	42.65	4-isopropyl-1,6-dimethylnaphthalene	$C_{15}H_{18}$	n.d.	4.36
15	44.90	3,6,9,12,15-pentaoxanonadecane-1,17-diol	$C_{14}H_{30}O_6$	9.70	4.85
16	45.34	2-(but-3-en-2-yl)-6-methoxynaphthalene	$C_{15}H_{16}O$	n.d.	3.73

related studies have to be done in the near future.

To further investigate the temperature effects on homogeneous reactions induced by CO_2, the temperature of the second furnace was increased to 700 °C. The syngas yields in all the pyrolysis setups (one- and two-stage at 500 °C and two-stage at 700 °C) are depicted in Fig. 7. The syngas formation is nearly negligible from one-stage pyrolysis, suggesting that additional thermal energy (≥500 °C) is required to produce syngas from POM. Notably, from the two-stage pyrolysis at 500 °C under CO_2 conditions, 30.47 mmol more syngas was produced than under N_2 conditions. This highlights the potential of using CO_2 as a reactant to optimize carbon/hydrogen utilization from POM through homogeneous reactions induced by CO_2. The two-stage pyrolysis at 700 °C showed enhanced syngas production under both N_2 and CO_2 conditions. However, the amount of syngas produced under CO_2 conditions was small (7 mmol). An excessive supply of thermal energy led to the complete decomposition of POM into syngas, making it difficult to observe the effect of CO_2 on the promotion of thermal cracking. Furthermore, high temperatures may consume excess energy to generate energy-related products (syngas). In conclusion, two-stage pyrolysis at 500 °C using CO_2 as the reactant is a strategic approach for the

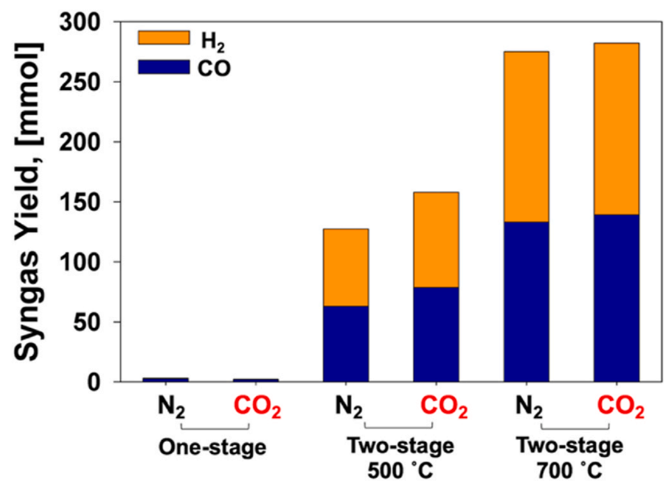

Fig. 7. H_2 and CO yields obtained from different pyrolysis setups (one- and two-stage at 500 °C and two-stage at 700 °C).

valorization of POM. This study did not use the catalyst in the pyrolysis process. To build a more energy-efficient plastic recycling platform, it is of importance to apply catalysts that can enable a reaction with lower activation energies during pyrolysis process. Further studies are required to develop catalysts that operate at lower temperatures and to investigate the effect of catalyst loading on CO_2-mediated pyrolysis of POM.

Syngas is important building blocks used for synthesizing value-added products (olefins, paraffins, alcohols, *etc.*) [30,31,56]. Product selectivity depends on the H_2 and CO ratio, which can be controlled via a widely established chemical process (*i.e.*, (reverse) water gas shift reaction) [57]. Thus, it is of importance to improve the production of syngas from the pyrolysis of POM. In other words, use of CO_2 in pyrolysis of POM leads to a substantial increase of syngas, compared to N_2 conditions. Considering this, it could be concluded that use of CO_2 in pyrolysis can be an economic feasible option to valorize the POM. In addition to this, use of CO_2 as a reactant in pyrolysis offers environmental benefits in that it can reduce the concentration of greenhouse gases in the atmosphere, contributing to alleviating climate changes.

4. Conclusions

In this study, pyrolysis is proposed as an environmentally benign platform for the valorization and disposal of plastic waste, specifically to convert it into syngas. A unique aspect of this study is the adoption of a pyrolysis process using CO_2 as the strategic reaction medium. The case study focused on POM, a widely used engineering plastic. Before initiating POM pyrolysis, the polymer type was identified using FT-IR spectrometry and elemental analysis, which revealed that the POM sample consisted of a copolymer-type POM with nitrogen-containing additives. The one-stage pyrolysis confirmed that CO_2 expedited the thermolysis of the POM liquid phase through a heterogeneous reaction. In the two-stage pyrolysis, the role of CO_2 in the heterogeneous reaction was substantially enhanced in the presence of homogeneous reactions. Notably, CO_2 expedited the reaction kinetics of dehydrogenation and deoxygenation, resulting in the production of 30.47 mmol more syngas compared to that from inert conditions. The study concluded that two-stage pyrolysis at 500 °C using CO_2 as a reactant represents a strategic approach for the valorization of POM. This approach demonstrates the potential of utilizing pyrolysis with CO_2 to convert plastic waste into valuable syngas, thereby contributing to environmental conservation and resource recovery. To build a more energy-efficient plastic recycling platform, further studies are required to apply catalysts for CO_2-mediated pyrolysis of POM in the future.

CRediT authorship contribution statement

Dohee Kwon: Writing – original draft, Data curation. **Dongho Choi:** Investigation, Formal analysis. **Hocheol Song:** Investigation, Formal analysis, Data curation. **Jechan Lee:** Supervision, Formal analysis. **Sungyup Jung:** Writing – review & editing, Formal analysis, Data curation. **Eilhann E. Kwon:** Writing – review & editing, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgements

This study was supported by a National Research Foundation of Korea (NRF) grant funded by the Korean government (MSIT) (Project No. RS-2023-00219667). Dohee Kwon acknowledges the financial support from the Hyundai Motor Chung Mong-Koo Foundation.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.energy.2024.132118.

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